Aditya Saini

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Education

Purdue University West Lafayette, IN
B.S. Computer Science (Systems Programming) *Graduation: May 2028 (Expected)*
Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Programming in C, Object-Oriented Programming, Discrete Mathematics, Linear Algebra, Multivariate Calculus

Experience

Embedded Engineer Sep. 2025 – Present
Purdue Electric Racing *West Lafayette, IN*

- Designed, developed, and deployed production C software for a real-world electric race car, writing modular RTOS firmware with unit-tested drivers and in-circuit debugging
- Built runtime telemetry and data acquisition to profile performance bottlenecks and improve reliability under changing race-day requirements
- Owned I2C/SPI HAL infrastructure APIs used by the team for high-frequency sensor communication, collaborating across software and electrical engineers
- Developed UART drivers and optimized command encoding for a real-time dashboard, debugging firmware/hardware interactions end-to-end

Chip Design Member (SoCET STARS) June 2026 – August 2026
SoCET, Purdue VIP *West Lafayette, IN*

- Applied computer architecture fundamentals by implementing MMIO peripheral drivers and accelerator integration for a custom RISC-V SoC
- Designed sequential/combinational logic in SystemVerilog and wrote self-checking unit testbenches, validating each block through the RTL-to-GDSII flow
- Wrote RISC-V assembly for the AFTx07 ISA and managed Linux builds with Makefiles and Git in a collaborative, requirements-driven design environment

Robotic Safety Research Assistant January 2026 – July 2026
PurSec, Purdue Computer Science *West Lafayette, IN*

- Researched jailbreak vulnerabilities in multimodal-LLM-controlled robots, independently scoping multi-step attack scenarios against state-of-the-art safety systems
- Built automated Python test pipelines on headless Linux with Docker and AWS EC2, running reproducible regression evaluations and metrics across 10+ scenarios

Projects

FRC Team 2554 | *Java, OpenCV, Jetson Orin Nano, CUDA* September 2021 – August 2025

- Led 40+ students as Build President through a dynamic competition season, coordinating software, mechanical, and electrical teams to reach District Championships after 18+ years.
- Developed and deployed the team's first Swerve Drive software stack, tuning PID/FF control and vision pipelines to improve autonomous positioning accuracy by 66% on a real competition robot.

Natural Language Shell | *C, Python, OpenAI API, Git* June 2025 – January 2026

- Built a C command-line shell with tokenizer, intent parsing, and bash mapping; integrated Python/OpenAI for complex queries, applying data structures and debugging across the full stack.

Tava | *React Native, TypeScript, JavaScript, Firebase* September 2025

- Shipped a cross-platform app connecting student builders via AI-powered search and swipe matching, with Firebase auth/data and interactive network visualization.

Technical Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, Rust, SQL, SystemVerilog
Systems: Computer Architecture, Networking (TCP/IP), Operating Systems, Concurrency, RTOS, Embedded Linux, performance optimization
Tools: Git, Docker, Linux, Makefiles, GDB, AWS EC2, unit/integration testing, debugging, CI-style regression pipelines
Hardware-Adjacent: STM32, SPI, I2C, UART, CAN, device drivers, HALs, in-circuit debugging